

MOHAMED KHALID ELBAHLOOL

Computer Engineering Senior — Embedded Systems & Software Platforms

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SUMMARY

Computer Engineering senior (B.S. expected Aug 2027) who designs, ships, and operates complete systems: bare-metal firmware debugged on real hardware, FPGA logic verified exhaustively in CI, and a self-hosted customer-service platform in production for a retail business. Trilingual — Arabic, English, Turkish. Every claim below is documented and traceable at www.mohamedelbahlool.com.

EXPERIENCE

Digital Systems & Automation Specialist — English Home Libya (Remote)

2023 – Present

- Run the digital operations of a retail brand serving customers in Libya: Arabic/English customer messaging, product catalogs, domain/DNS, and branded email for a Facebook page with ~61,000 followers (public page count, July 2026).
- Designed, built, and operate a self-hosted customer-service platform (TypeScript, Next.js, PostgreSQL, Docker Compose, Caddy, Gemini API, Meta Graph API) answering product inquiries in Libyan Arabic — catalog-true prices across 4,700+ products, photo-based product recognition, and instant human handoff.
- Migrated the platform off a paused free-tier database host to a self-hosted VPS with nightly backups; added safety mechanisms (burst batching, supersede guard, send-time re-checks, output sanitizer) that each trace to a real production incident.
- Built the catalog pipeline: a scraper that assembled the full catalog (image set exceeding 11 GB during the documented build) plus a trilingual (AR/EN/TR) product-matching engine.

Learning Assistant II — FIU Mastery Math Lab, Florida International University

Aug 2024 – Present

- Support Algebra, Precalculus, and Trigonometry students ~20 hours/week through mini-lessons, guided practice, and one-on-one tutoring — daily practice explaining technical ideas to students from very different backgrounds.

PROJECTS

NOVA RAID — bare-metal arcade game, Raspberry Pi Pico W C · RP2040 · Pico SDK · 2026 · github.com/lla7wel/nova-raid

- Complete retro space shooter written bare-metal, solo: dual-core 25 fps render pipeline, SPI at 62.5 MHz with double-buffered DMA, 24.8 fixed-point math; testable HAL with a desktop host harness; released as drag-and-drop UF2s with green CI.
- First run on real hardware exposed a display-init defect no simulator caught — diagnosed, fixed, and documented same day in v1.0.1.

Pico W voice terminal & self-hosted control plane C · PIO/DMA · Python · FastAPI · 2026 · github.com/lla7wel/Pico-AI-Agent-System

- Hand-wired push-to-talk voice assistant: SPH0645 I2S microphone captured via a custom 8-instruction PIO program with a DMA ring buffer, streaming over one long-lived TLS WebSocket to a self-hosted FastAPI control plane with permission-gated tools; verified end-to-end by a fully offline integration test in CI.

Verified digital logic on FPGA VHDL · GHDL · Vivado · Basys 3 · Summer 2026 · github.com/lla7wel/fpga-ripple-carry-adder

- Coursework rebuilt as real releases: exhaustive self-checking testbenches (512/512 adder vectors, 36/36 decoder vectors) in GHDL simulation on every push, with Vivado waveforms and Basys 3 pin constraints.

Smart-home appliance controller C++ · Arduino UNO · two-person team (with Paola Dorado Galicia) · Spring 2026

- DHT11 sensing, PWM fan control, four-mode state machine, 35 °C over-temperature cutoff; maintained the repository, documentation, and schematics.

Miramar — bilingual EN/TR business website Next.js 16 · React 19 · TypeScript · live at tr-miramar.com

- Designed, built, and deployed solo for an Istanbul construction-materials supplier; Playwright smoke suite green in CI.

SKILLS

Languages: C, C++, Python, TypeScript/JavaScript, VHDL, SQL, HTML/CSS

Embedded: RP2040 / Pico SDK, PIO, DMA, SPI/I2S/UART, Arduino, sensor integration, fixed-point math

Systems & web: Next.js, React, Astro, FastAPI, PostgreSQL, Docker Compose, Caddy, REST/webhooks/WebSocket, Meta Graph API, DNS

Verification: GitHub Actions CI, Playwright, GHDL simulation, Vivado, Git/GitHub · Human languages: Arabic, English, Turkish

EDUCATION

Florida International University — B.S. Computer Engineering · GPA 3.6 · Dean's List Fall 2024, Spring 2025 · senior (93 credits) Expected Aug 2027